CHAPTER 7

Redox Reactions

Redox Reactions (Oxidation and Reduction), Oxidation Number

1. The oxidation states not shown by Mn in given reaction is:

$$3MnO_4^{2-} + 4H^+ \longrightarrow 2MnO_4^- + MnO_2 + 2H_2O$$

A. +6 B. +2 C. +4 D. +7 E. +3

Choose the most appropriate answer from the options given below : *(2024 Re)*

a. D and E only b. B and D only
c. A and B only d. B and E only

2. Which reaction is **NOT** a redox reaction? *(2024)*

a. $H_2 + Cl_2 \rightarrow 2\,HCl$
b. $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2\,NaCl$
c. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
d. $2\,KClO_3 + I_2 \rightarrow 2\,KIO_3 + Cl_2$

3. The **correct** option for a redox couple is: *(2023-Manipur)*

a. Both are oxidised forms involving same element.
b. Both are reduced forms involving same element.
c. Both the reduced and oxidized forms involve same element.
d. Cathode and anode together.

4. What is the change in oxidation number of carbon in the following reaction? *(2020)*

$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(l) + 4HCl(g)$

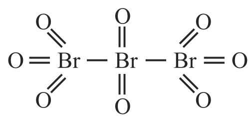
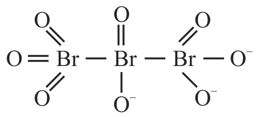
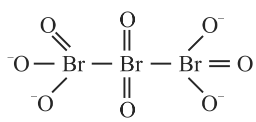
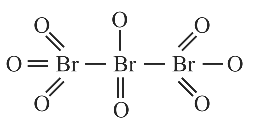
a. 0 to + 4 b. –4 to + 4
c. 0 to –4 d. +4 to + 4

5. The oxidation number of the underlined atom in the following species *(2020-Covid)*

a. $\underline{Cl}O_3^-$ is +5 b. $K_2\underline{Cr_2}O_7$ is +6
c. $H\underline{Au}Cl_4$ is +3 d. $Cu_2\underline{O}$ is –1

Identify the incorrect option

6. The correct structure of tribromooctaoxide is *(2019)*

a. 
b. 
c. 
d. 

7. Hot concentrated sulphuric acid is a moderately strong oxidising agent. Which of the following reactions does not show oxidising behaviour? *(2016 - I)*

a. $C + 2H_2SO_4 \rightarrow CO_2 + 2SO_2 + 2H_2O$
b. $CaF_2 + H_2SO_4 \rightarrow CaSO_4 + 2HF$
c. $Cu + 2H_2SO_4 \rightarrow CuSO_4 + SO_2 + 2H_2O$
d. $3S + 2H_2SO_4 \rightarrow 3SO_2 + 2H_2O$

8. In acidic medium, H_2O_2 changes $Cr_2O_7^{-2}$ to CrO_5 which has two **(—O—O—)** bonds. Oxidation state of Cr in CrO_5 is: *(2014)*

a. +3 b. +6 c. –10 d. +5

9. The pair of compounds that can exist together is *(2014)*

a. $FeCl_3$, $SnCl_2$ b. $HgCl_2$, $SnCl_2$
c. $FeCl_2$, $SnCl_2$ d. $FeCl_3$, KI

10. The oxidation state of Cr in CrO_5 is *(2014)*

a. –6 b. +12 c. +6 d. +4

Types of Redox Reactions and Balancing of Redox Reactions

11. On balancing the given redox reaction, *(2023)*

$aCr_2O_7^{(2-)} + bSO_3^{(2-)}(aq) + cH^+(aq) \rightarrow$
$2aCr^{3+}(aq) + bSO_4^{2-}(aq) + c/2\,H_2O\,(l)$

the coefficients a, b and c are found to be, respectively:

a. 8, 1, 3 b. 1, 3, 8 c. 3, 8, 1 d. 1, 8, 3

12. Which of the following reactions is a decomposition redox reaction? *(2022 Re)*

a. $P_4(s) + 3OH^-(aq) + 3H_2O(l) \rightarrow PH_3(g) + 3H_2PO_2^-(aq)$
b. $2\,Pb(NO_3)_2(s) \rightarrow 2PbO(s) + 4NO_2(g) + O_2(g)$
c. $N_2(g) + O_2(g) \rightarrow 2NO(g)$
d. $Cl_2(g) + 2OH^-(aq) \rightarrow ClO^-(aq) + Cl^-(aq) + 4H_2O(l)$

13. Which of the following reactions is the metal displacement reaction? Choose the right option. *(2021)*

a. $Cr_2O_3 + 2Al \xrightarrow{\Delta} Al_2O_3 + 2Cr$
b. $Fe + 2HCl \longrightarrow FeCl_2 + H_2\uparrow$
c. $2Pb(NO_3)_2 \longrightarrow 2PbO + 4NO_2 + O_2\uparrow$
d. $2KClO_3 \xrightarrow{\Delta} 2KCl + 3O_2$

14. Which of the following reactions are disproportionation reaction? *(2019)*

A. $2Cu^{+} \longrightarrow Cu^{2+} + Cu^{0}$

B. $3MnO_4^{2-} + 4H^{+} \longrightarrow 2MnO^{-}{}_4 + MnO_2 + 2H_2O$

C. $2KMnO_4 \xrightarrow{\Delta} K_2MnO_4 + MnO_2 + O_2$

D. $2MnO_4^{-} + 3Mn^{2+} + 2H_2O \longrightarrow 5MnO_2 + 4H^{\oplus}$

Select the correct option from the following

a. (A) and (B) only
b. (A), (B) and (C)
c. (A), (C) and (D)
d. (A) and (D) only

15. For the redox reaction

$$MnO_4^{-} + C_2O_4^{2-} + H^{+} \rightarrow Mn^{2+} + CO_2 + H_2O$$

The correct coefficients of the reactants for the balanced equation are: *(2018)*

	MnO_4^-	$C_2O_4^{2-}$	H^+
a.	16	5	2
b.	2	5	16
c.	5	16	2
d.	2	16	5

Answer Key

1. (d)	**2.** (b)	**3.** (c)	**4.** (b)	**5.** (d)	**6.** (a)	**7.** (b)	**8.** (b)	**9.** (c)	**10.** (c)
11. (b)	**12.** (b)	**13.** (a)	**14.** (a)	**15.** (b)					